

A	P1	If I was reading the newspaper in the kitchen, then my glasses are on the kitchen table.
B	P2	If my glasses are on the kitchen table, then I saw them at breakfast.
C	P3	I did not see my glasses at breakfast.
D	P4	I was reading the newspaper in the living room or I was reading the newspaper in the kitchen.
E	P5	If I was reading the newspaper in the living room then my glasses are on the coffee table.

Lets assume that

Reading the newspaper in the kitchen.	RK
Glasses are on the kitchen table	GK
I saw my glasses at breakfast.	SB
Reading the newspaper in the living room.	RL
Glasses are on the coffee table.	GC

Apply Inference Rules by using these premises

MODUS PONENS RULE

Rule Name	Order	General form
Modus Ponens Rules	$RL \rightarrow GC$ (by C) RL (by 3) $\therefore GC$ (by modus ponens)	$P \rightarrow Q$ P $\therefore Q$ (by modus ponens)
P = RL = Reading the newspaper in the living room. Q = GC = Glasses are on the coffee table. C = RL = Reading the newspaper in the living room.		
In NLP above premises will be P $\rightarrow$ Q = If Reading the newspaper in the living room Then Glasses are on the coffee table. P = Glasses are on the coffee table.		
In this case, we are given the premises: 1. $RL \rightarrow GC$ (Reading the newspaper in the living room implies glasses are on the coffee table) 2. RL (Reading the newspaper in the living room) From premise 1, we have an implication where RL (Reading the newspaper in the living room) is the antecedent and GC (glasses are on the coffee table) is the consequent. From premise 2, we know that RL (Reading the newspaper in the living room) is true. Using Modus Ponens, we can conclude that GC (Glasses are on the coffee table) must be true. Therefore, the conclusion is Q (Glasses are on the coffee table). Please note that the given information does not provide enough context to determine the specific order or form of the Modus Ponens rule being used. The general form of Modus Ponens is typically represented as $P \rightarrow Q, P, \text{therefore } Q$.		

MODUS TOLLENS RULE

Rule Name	Order	General form
Modus tollens Rules	$RK \rightarrow SB$ $\sim SB$ $\therefore \sim RK$ (by modus tollens)	$P \rightarrow Q$ $\sim Q$ $\therefore \sim P$ (by modus ponens)
P = RK = Reading the newspaper in the kitchen. Q = SB = i saw my glasses at breakfast $\sim Q = \sim SB$ = i did not saw my glasses at breakfast C = $\sim RK$ = I am not Reading the newspaper in the kitchen.		
In NLP above premises will be P $\rightarrow$ Q = If Reading the newspaper in the kitchen Then i saw my glasses at breakfast $\sim Q$ = i not saw my glasses at breakfast		
In this case, we are given the premises: 1. $RK \rightarrow SB$ (Reading the newspaper in the kitchen implies seeing glasses at breakfast) 2. $\sim SB$ (I did not see my glasses at breakfast) From premise 1, we have an implication where RK (Reading the newspaper in the kitchen) is the antecedent and SB (seeing glasses at breakfast) is the consequent. From premise 2, we know that SB (seeing glasses at breakfast) is false ($\sim SB$). Using Modus Tollens, we can conclude that $\sim RK$ (I am not reading the newspaper in the kitchen) must be false. Therefore, the conclusion is $\sim P$ (I am not reading the newspaper in the kitchen). Please note that the given information does not provide enough context to determine the specific order or form of the Modus Ponens rule being used. The general form of Modus Tollens is typically represented as $P \rightarrow Q, \sim Q$, therefore $\sim P$.		

ELIMINATION RULE

Rule Name	Order	General form
Elimination rule	$RL \vee RK$ $\sim RK$ $\therefore RL$	$P \vee Q$ $\sim Q$ $\therefore P$ (by elimination rule)
P = RL = Reading the newspaper in the living room Q = RK = Reading the newspaper in the kitchen $\sim Q = \sim RK$ = i am not Reading the newspaper in the kitchen C = RL = Reading the newspaper in the living room		
In NLP above premises will be P $\vee$ Q = either Reading the newspaper in the living room or reading the newspaper in the kitchen $\sim Q$ = i am not Reading the newspaper in the kitchen		
In this case, we are given the premises: 1. $P \vee Q$ (Reading the newspaper in the living room or Reading the newspaper in the kitchen) 2. $\sim Q$ (I am not reading the newspaper in the kitchen)		

From premise 1, we know that at least one of the disjuncts must be true. Since we know that Q is false ($\sim Q$), we can conclude that P (Reading the newspaper in the living room) must be true.

Therefore, the conclusion is P (Reading the newspaper in the living room).

TRANSITIVITY RULE

Rule Name	Order	General form
Transitivity rule	$RK \rightarrow GK$ (by A) $GK \rightarrow SB$ (by D) $\therefore RK \rightarrow SB$ (Transitivity)	$P \rightarrow Q$ $Q \rightarrow R$ $\therefore P \rightarrow R$
P = RK = Reading the newspaper in the kitchen Q = GK = Glasses are on the kitchen table R = SB = I saw my glasses at breakfast.		
In NLP above premises will be P $\rightarrow$ Q = IF newspaper in the kitchen THEN Glasses are on the kitchen table Q $\rightarrow$ R = IF Glasses are on the kitchen table THEN I saw my glasses at breakfast. P $\rightarrow$ R = IF newspaper in the kitchen THEN I saw my glasses at breakfast.		
In this case, we are given the premises: 1. $RK \rightarrow GK$ (If reading the newspaper in the kitchen, then glasses are on the kitchen table) 2. $GK \rightarrow SB$ (If glasses are on the kitchen table, then I saw my glasses at breakfast) From premise 1 , we have an implication where RK (reading the newspaper in the kitchen) is the antecedent and GK (glasses are on the kitchen table) is the consequent. From premise 2 , we have another implication where GK (glasses are on the kitchen table) is the antecedent and SB (I saw my glasses at breakfast) is the consequent. Using the Transitivity rule, we can combine the two implications to conclude that RK (reading the newspaper in the kitchen) implies SB (I saw my glasses at breakfast), resulting in the implication $P \rightarrow R$ (If newspaper in the kitchen, then I saw my glasses at breakfast). Therefore, the conclusion is $P \rightarrow R$ (If newspaper in the kitchen, then I saw my glasses at breakfast).		